

PHASE 3 PRESENTATION

Detailed Design & Execution Plan

Automated Resume Screening: Ethical AI

Proposed Solution Features



Contextual NLP

Moving beyond keyword matching to capture semantic intent via Transformers.



Fairness-Aware

Integrated DIR/DPD monitoring to ensure EEOC and regulatory compliance.



Modular Pipeline

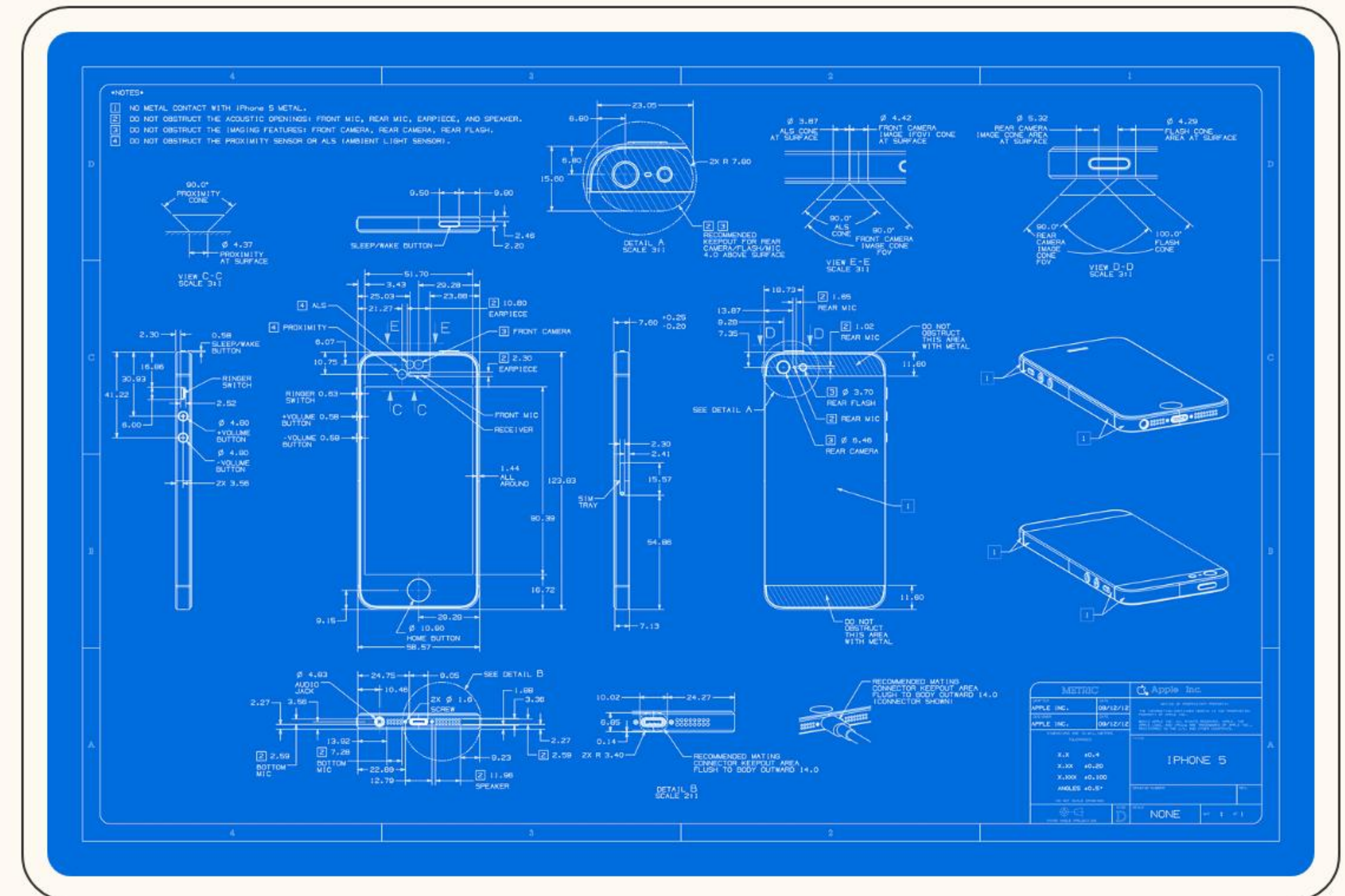
Decoupled preprocessing and mitigation modules for enterprise scalability.

Design Selection: Dual-Track

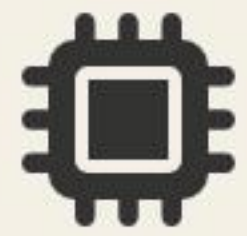
Baseline Track: Uses TF-IDF and Classical ML (LR/SVM) for high interpretability and speed.

Advanced Track: Uses BERT/RoBERTa for deep contextual understanding of skills and experience.

Merger: The system selects the most accurate model while applying a fairness-reweighting layer.



Identification of Constraints



Computational

GPU requirements for Transformer fine-tuning (BERT-base vs Large).



Data Privacy

Handling PII (Personally Identifiable Information) in resume datasets.



Ethical

Balancing classification accuracy with demographic neutrality.

Feature Finalization Matrix

Feature Category	Selected Implementation	Constraint Analysis
Embeddings	TF-IDF & RoBERTa	Memory vs Context Trade-off
Bias Detection	DIR, DPD, EOD	Quantitative Standard (EEOC)
Mitigation	Adversarial Debiasing	Training Complexity (In-processing)

Feasibility & Budgeting

100%

TECHNICAL FEASIBILITY

Low

OPERATIONAL BUDGET

System utilizes Open Source frameworks (PyTorch, spaCy, Transformers) ensuring cost-efficiency.

Regulatory Standards

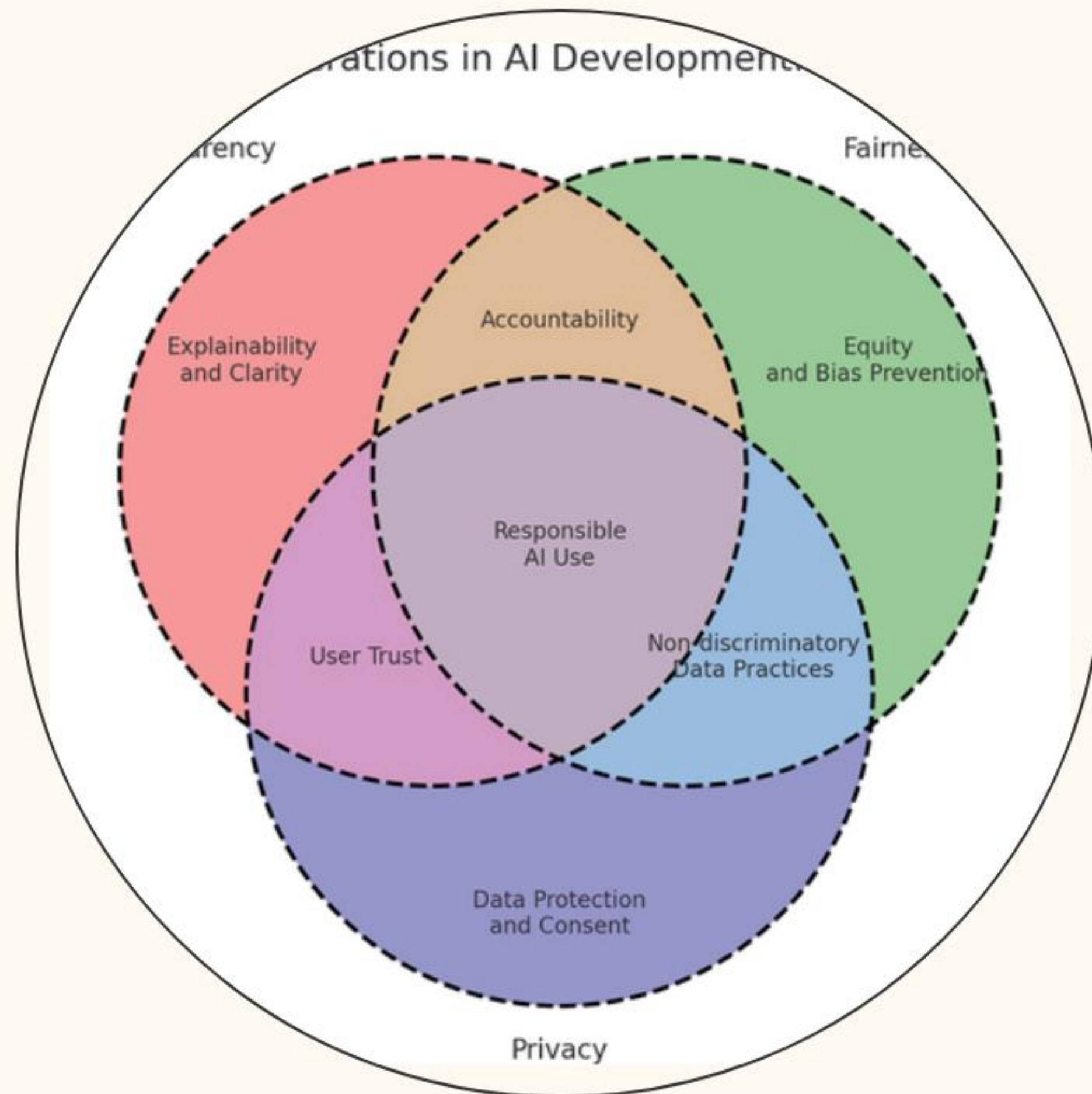


- **EEOC Compliance:** Adhering to the "Four-Fifths Rule" ($DIR > 0.80$).
- **GDPR Guidelines:** Right to explanation in automated decisions.
- **Ethical AI Principles:** Transparency and accountability in recruitment.

Execution Plan Roadmap



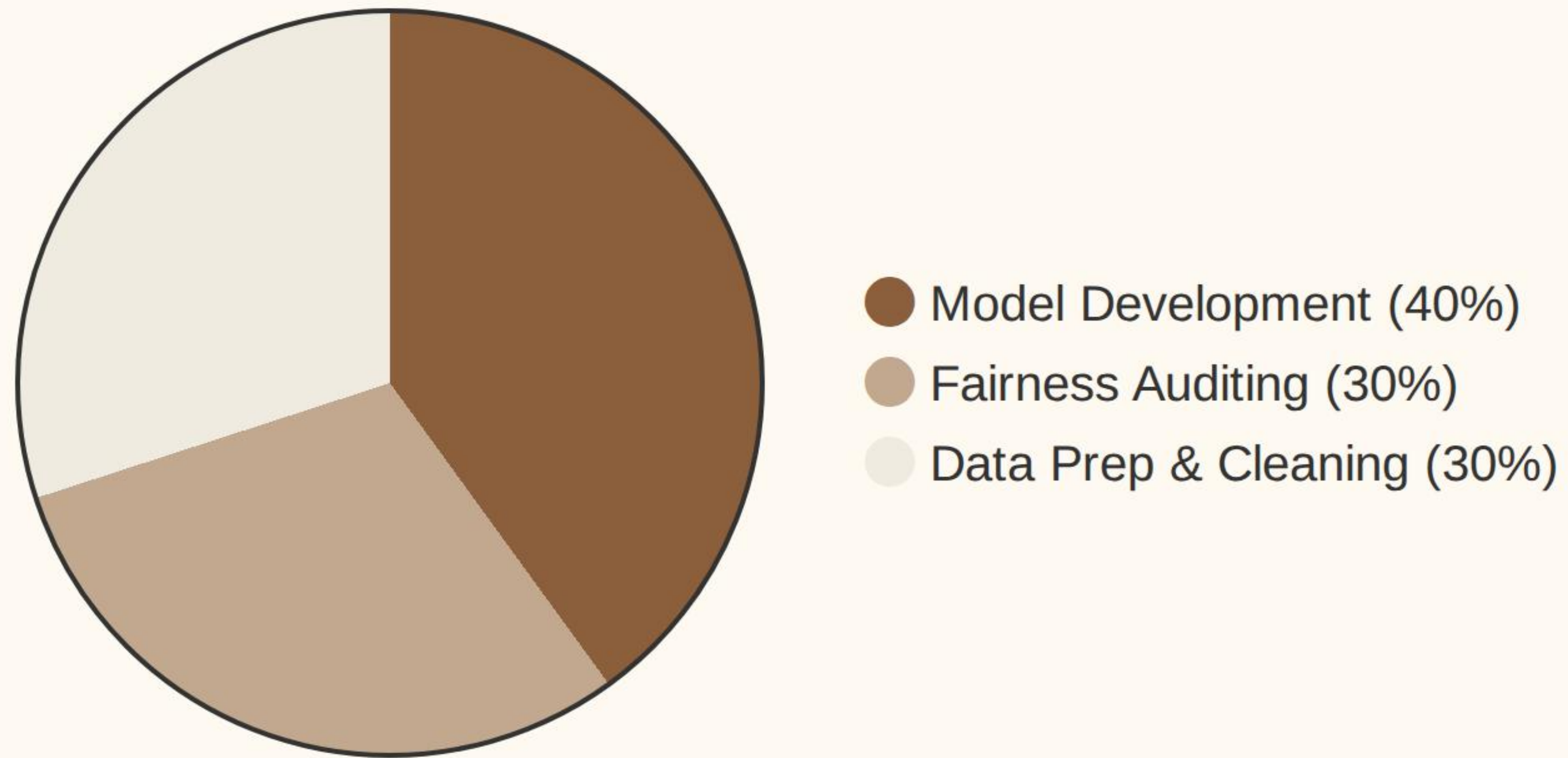
Bias Mitigation Strategy



Three-Tier Mitigation

1. **Pre-processing:** Sample re-weighting.
2. **In-processing:** Adversarial Debiasing.
3. **Post-processing:** Threshold optimization.

Resource Allocation



"Efficiency in recruitment must not overwrite equity."

Q&A

END OF PHASE 3

Image Sources



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